

Seminar 2 - LL(1)

$\overline{LL(k)}$ uses leftmost derivations
↑ predictions of length k
↑ sequence read from left to right

LL(k) parser: at each moment of parsing an action is uniquely determined

LL(1) Parser

→ uses predictions of length 1

Steps

- 1) build FIRST & FOLLOW
- 2) build LL(1) parse table
- 3) analyse sequence based on moves and table

FIRST

let $\alpha \in (N \cup \Sigma)^*$, then $FIRST_k$ means first k terminal symbols that can be generated by α

Notation $FIRST_1 = FIRST$

Example:

$S \rightarrow Bb \mid Cd$

$B \rightarrow aB \mid \epsilon$

$C \rightarrow cC \mid \epsilon$

	F_0	F_1	F_2
S	$\emptyset$	a, ϵ, c, d	a, ϵ, c, d
B	a, ϵ	a, ϵ	a, ϵ
C	c, ϵ	c, ϵ	c, ϵ

Ex1) Compute FIRST for:

$S \rightarrow BA^1$
 $A \rightarrow +BA^2$
 $A \rightarrow \epsilon^3$
 $B \rightarrow DC^4$
 $C \rightarrow *DC^5$
 $C \rightarrow \epsilon^6$
 $D \rightarrow (S)^7$
 $D \rightarrow a^8$

	F ₀	F ₁	F ₂	F ₃
S	$\emptyset$	$\emptyset$	(, a	(, a
A	+ , ϵ	+ , ϵ	+ , ϵ	+ , ϵ
B	$\emptyset$	(, a	(, a	(, a
C	*, ϵ	*, ϵ	*, ϵ	*, ϵ
D	(, a	(, a	(, a	(, a

FIRST(S) = { (, a }

FIRST(A) = { + , ϵ }

FIRST(B) = { (, a }

FIRST(C) = { *, ϵ }

FIRST(D) = { (, a }

FOLLOW

→ the next symbol generated after/following A

Rules (FOLLOW(A) for example)

we look for A in the right side of all productions

- if A is at the end of production we add (union) FOLLOW (left side of prod.)

- if after A is a terminal, that terminal is the FOLLOW

- if after A is another non-terminal, let's say B, then FOLLOW(A) = FIRST(B) and if FIRST(B) contains ϵ then we also add FOLLOW (left-side) !

example

$S \rightarrow Bb | Cd$

$B \rightarrow aB | \epsilon$

$C \rightarrow cC | \epsilon$

	L ₀	L ₁	L ₂
S	ϵ	ϵ	ϵ
B	$\emptyset$	b	b
C	$\emptyset$	d	d

Algorithm FOLLOW

INPUT: $G, \text{FIRST}(X), \forall X \in N \cup \Sigma$

OUTPUT: $\text{FOLLOW}(A), \forall A \in N$

for $A \in N - \{S\}$ **do**

{init}

$L_0(A) = \Phi;$

endFor;

$L_0(S) = \{\epsilon\};$

{init}

$i = \underline{0};$

repeat

$i = i + \underline{1};$

for $B \in N$ **do**

for $A \rightarrow \alpha B \gamma \in P$ **do**

for $\forall a \in \text{FIRST}(\gamma)$ **do**

if $a = \epsilon$ **then** $F_i(B) = F_i(B) \cup F_{i-1}(A)$

else $F_i(B) = F_{i-1}(B) \cup \text{First}(\gamma)$

endif

endFor

endFor

endfor

until $F_i(X) = F_{i-1}(X), \forall X \in N$

$\text{FOLLOW}(X) = F_i(X), \forall X \in N$

Rule for L_0

$$L_0(S) = \{\epsilon\}$$

$$L_0(x) = \emptyset \quad \forall x \in N - \{S\}$$

Ex 2

Compute FOLLOW for the grammar from EX 1

	L_0	L_1	L_2	L_3	L_4
S	ϵ	$\epsilon,)$	$\epsilon,)$	$\epsilon,)$	$\epsilon,)$
A	$\emptyset$	ϵ	$\epsilon,)$	$\epsilon,)$	$\epsilon,)$
B	$\emptyset$	$+, \epsilon$	$+, \epsilon,)$	$+, \epsilon,)$	$+, \epsilon,)$
C	$\emptyset$	$\emptyset$	$+, \epsilon$	$+, \epsilon,)$	$+, \epsilon,)$
D	$\emptyset$	$*, \epsilon$	$*, \epsilon, +$	$*, \epsilon, +,)$	$*, \epsilon, +,)$

$$FOLL(S) = \{ \epsilon \}$$

$$FOLL(A) = FOLL(S) \cup FOLL(A)$$

$$FOLL(B) = FIRST(A) \cup FOLL(S) \cup FOLL(B)$$

because $FIRST(A)$ contains ϵ

$$FOLL(C) = FOLL(B) \cup FOLL(C)$$

$$FOLL(D) = FIRST(C) \cup FOLL(B) \cup FOLL(C)$$

because $FIRST(C)$ contains ϵ

LL(1) TABLE

$M(A, a) = (\alpha, i) \quad \forall a \in \text{FIRST}(\alpha), a \neq \epsilon, A \rightarrow \alpha$
production with number i

$M(A, b) = (\alpha, i) \quad \text{if } \epsilon \in \text{FIRST}(\alpha), \forall b \in \text{FOLLOW}(A),$
 $A \rightarrow \alpha$ production with number i

$M(a, a) = \text{pop}, \forall a \in \Sigma \quad (\Rightarrow \text{terminal} + \text{terminal} = \text{pop})$

$M(\$, \$) = \text{acc}$

$M(x, a) = \text{err}$

example: for example FIRST & example FOLLOW

	a	b	c	d	\$
S	Bb, 1	Bb, 1	cd, 2	cd, 2	
B	ab, 3	ϵ , 4			
C			cc, 5	ϵ , 6	
a	pop				
b		pop			
c			pop		
d				pop	
\$					acc

conf (α, β, π)
 input stack $\nearrow$ $\uparrow$ $\nwarrow$ output
 workin stack

init: $(w \$, S \$, \epsilon)$

end: $(\$, \$, \pi)$

$w = aab \in L(G)?$

$(\underline{a}ab \$, \underline{S} \$, \epsilon) \xrightarrow{\text{push}} (\underline{a}ab \$, \underline{Bb} \$, 1) \xrightarrow{\text{push}} (\underline{aa} \$, \underline{aBb} \$, 13)$
 $\xrightarrow{\text{pop}} (\underline{ab} \$, \underline{Bb} \$, 13) \xrightarrow{\text{push}} (\underline{ab} \$, \underline{aBb} \$, 133) \xrightarrow{\text{pop}} (\underline{b} \$, \underline{Bb} \$, 133)$
 $\xrightarrow{\text{pop}} (\underline{b} \$, \underline{\epsilon b} \$, 1334) \xrightarrow{\text{pop}} (\underline{\$} \$, \underline{\$} \$, 1334) \quad \checkmark$

Ex 3

Build the LL(1) table for ex 1 & 2

	+	*	(	)	a	\$
S			BA,1		BA,1	
A	+BA,2			E,3		
B			DC,4		DC,4	
C	E,6	*DC,5		E,6		
D			(S),7		a,8	
+	pop					
*		pop				
(			pop			
)				pop		
a					pop	
\$						acc

$w = a * (a + a)$? $L(G)$

$$D \rightarrow a^8$$

$\triangleright (a^*(a+a)\$, S\$, \epsilon) \vdash (a^*(a+a)\$, BA\$, 1) \vdash (a^*(a+a)\$, DCA\$, 14)$
 $\vdash (a^*(a+a)\$, aCA\$, 148) \vdash (* (a+a)\$, CA\$, 148) \vdash ((* (a+a)\$, *DCA\$, 1485)$
 $\vdash ((a + a)\$, DCA\$, 1485) \vdash ((a+a)\$, (S)CA\$, 14857)$
 $\vdash (a+a)\$, (S)CA\$, 14857) \vdash (a+a)\$, BA)CA\$, 148571) \vdash (a+a)\$, DCA)CA\$,$
 $1485714) \vdash (a+a)\$, aCA)CA\$, 14857148) \vdash (+a)\$, CA)CA\$, 14857148)$
 $\vdash (+a)\$, \epsilon A)CA\$, 148571486) \vdash (+a)\$, +BA)CA\$, 1485714862) \vdash (a)\$,$
 $BA)CA\$, 1485714862) \vdash (a)\$, DCA)CA\$, 14857148624) \vdash (a)\$,$
 $aCA)CA\$, 148571486248) \vdash ()\$, CA)CA\$, 148571486248) \vdash ()\$,$
 $\epsilon A)CA\$, 1485714862486) \vdash ()\$, \epsilon)CA\$, 14857148624863) \vdash (\$, CA\$,$
 $14857148624863) \vdash (\$, \epsilon A\$, 148571486248636) \vdash (\$, \epsilon\$,$
 $1485714862486363) \vdash (\$, \$\$, 1485714862486363) \vdash acc$